

Primum Non Nocere? Mortality Effects of a Physician Strike in Germany

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Abstract

We estimate the causal effect of a protracted, unanticipated physician strike in German university hospitals on patient mortality. Using a difference-in-differences design and a double-selection LASSO algorithm, we show that the strike sharply decreased admissions and increased in-hospital mortality by 9 percent. Altered patient composition and strategic triaging account for half of this mortality spike, while the remainder likely reflects diminished care quality. Nearby non-striking hospitals experienced patient spillovers without elevated mortality, and out-of-hospital deaths remained unchanged. We conclude that prolonged healthcare strikes severely threaten patient safety unless strict risk-mitigation protocols are established.

Keywords: physician strike, quality of care, difference-in-differences, hospitals, spillover effects.

JEL Classifications: C21, I11, I18, J52.

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1 Introduction

Physician labour disruptions spark intense public debate because a doctor's professional duty to treat patients often conflicts with their fundamental right to strike ([Metcalf, Chowdhury and Salim, 2015](#)).¹ Even brief healthcare disruptions trigger severe concerns regarding patient safety, service continuity, and systemic resilience. Yet, despite heavy media coverage and the ethical controversies surrounding these walkouts, robust causal evidence evaluating the precise impacts of physician strikes on patient health remains scarce.

This paper provides new evidence on this topic by leveraging an unexpected nationwide physician strike in Germany to identify its causal effect on mortality. The three-month strike in 2006 followed a sudden breakdown in negotiations with the physicians' labour union. As one of the longest labour disruptions in German history, participation peaked at approximately two-thirds of physicians in tertiary hospitals.² This massive scale and unexpected onset create a unique natural experiment to isolate the impacts of a major disruption to healthcare supply.

The net effect of a physician strike on patient health is theoretically ambiguous and remains an empirical question. On the one hand, labour disruptions can degrade care quality by restricting access, intensifying staff workloads, and inducing crowding. For instance, [Awdic \(2016\)](#) demonstrates that increased travel distance from emergency department closures reduced short-run survival for acute myocardial infarctions due to elevated out-of-hospital mortality risks. Similarly, [Piérard \(2014\)](#), [Lin \(2014\)](#) and [Aiken et al. \(2002\)](#) underscore the critical role of staff-to-patient ratios in driving patient outcomes. Conversely, hospitals can adapt by prioritizing urgent cases and postponing elective procedures. This rationing could improve short-term outcomes by curbing unnecessary medical interventions. Given that hospital overtreatment is well-documented ([Lyu et al., 2017](#)), healthcare disruptions could plausibly yield neutral or even beneficial net effects.

¹The International Labour Organization (ILO) classifies hospitals as "essential services", meaning that strikes may be restricted or prohibited if they pose risks to life, safety, or public health (cf. [International Labour Office, 2006](#)).

²The majority of German tertiary hospitals are organised on federal state level and provide medical care to all publicly as well as privately insured individuals. They incorporate medical schools and are also involved in research and teaching activities.

We address this question by combining comprehensive administrative data on hospital admissions and mortality with newly collected hospital-level strike participation records. To construct this dataset, we digitized media reports to identify the exact timeline of the strike for each German university hospital. Our baseline difference-in-differences (DiD) specification compares patient outcomes at strike-affected hospitals with those at hospitals that never participated in the strike, before and after the strike. Crucially, non-participating hospitals were insulated from the strike not by choice, but because pre-existing governance structures placed them outside the collective bargaining unit. These historical institutional arrangements predated the strike and were unrelated to contemporaneous hospital performance, providing a plausibly exogenous source of variation in exposure to the strike.

Our baseline results show that the strike caused a 12% reduction in hospital admissions at affected facilities relative to controls, with no subsequent catch-up effects. Concurrently, the strike led to a significant 0.16 percentage point increase in in-hospital mortality, representing a 9% rise over the baseline. However, this baseline estimate may conflate care quality degradation with case-mix shifts if affected facilities prioritised more urgent or frail patients. To isolate the underlying mechanisms, we implement a double-selection LASSO model ([Belloni, Chernozhukov and Hansen, 2014](#)) to condition on a high-dimensional set of observable patient characteristics. After adjusting for these strike-induced composition shifts, the mortality coefficient decreases by half, but remains statistically significant. This attenuation confirms that the observed mortality spike stems jointly from strategic triaging and genuine reductions in care quality.

While the covariate adjustment accounts for strike-induced changes in the observable patient case-mix, it does not identify which specific characteristics drive these compositional shifts. Furthermore, the attenuation of the mortality coefficient establishes the aggregate contribution of patient composition to the overall mortality effect, but it leaves the precise underlying mechanisms unobserved. To disentangle these channels, we directly examine changes in individual patient characteristics and admission patterns. We show that patients admitted to striking hospitals during the strike were younger and more likely to require urgent

care. Consistent with this shift, emergency admissions increased by 5%, and the average length of stay rose by 0.2 days, reflecting the more intensive clinical needs of a systematically sicker patient population. While these shifts suggest that healthier individuals were less likely to receive hospital treatment, this did not translate into excess mortality outside the clinical setting. Utilizing comprehensive administrative records on all deaths in Germany, we find no evidence of increased out-of-hospital mortality, confirming that the adverse health consequences of the strike were largely confined to inpatient care.

Finally, we evaluate potential spillover effects on nearby non-striking hospitals, where displaced patients seeking alternative care could strain capacity and compromise care quality. We document modest increases in admissions at neighbouring facilities during the strike period, but detect no corresponding rise in in-hospital mortality. Taken together, these results imply that healthier patients at striking hospitals either neglected care or were successfully triaged, shifting the inpatient composition toward a higher share of high-risk, frail individuals. However, our case-mix-adjusted mortality estimates demonstrate that this care rationing and patient selection can only explain half of the observed mortality spike, with the remaining half pointing to diminished quality of care.

We contribute to the sparse literature on the causal effects of healthcare strikes. To our knowledge, only two prior studies examine physician-led walkouts. [Costa \(2022\)](#) investigates strikes across multiple medical staff categories in Portugal, finding that while nurse and technician strikes had no impact, physician strikes increased in-hospital mortality by 8%. [Stoye and Warner \(2023\)](#) analyse a series of brief junior doctor strikes in the United Kingdom in 2016. While conceptually related, our empirical context differs in three important respects. First, unlike the short, repeated disruptions examined in prior work, we study a protracted, three-month strike affecting entire university hospital systems. This represents a substantially more severe and sustained supply shock, while simultaneously allowing more time for institutional adaptation. Second, we quantify the relative contribution of changes in patient case-mix versus changes in care quality to the mortality effect, using LASSO-based methods. Third, our setting allows us to examine mortality displacement outside hospitals, which would

arise if deaths shifted from inpatient to out-of-hospital settings, and spillovers to unaffected hospitals, which could arise if patient reallocation created congestion at non-striking facilities. Together, these features allow us to provide a more complete picture of how healthcare systems respond to an extended, unanticipated institutional disruption.

Regarding labour disruptions among other healthcare professions, [Gruber and Kleiner \(2012\)](#) analyse multiple nursing strikes in New York, finding that these walkouts increased in-hospital mortality by 18% and degraded clinical quality. Similarly, [Kronborg, Sievertsen and Wüst \(2016\)](#) and [Hirani, Sievertsen and Wüst \(2019\)](#) evaluate a 2008 nurses' strike in Denmark, documenting reductions in prenatal midwife consultations, shorter post-partum hospitalizations, and fewer home visits. In a developing country context, [Friedman, Keats and Mutua \(2022\)](#) examine an all-encompassing health worker strike in Kenya, finding that infants born during strike months faced an elevated risk of mortality. We extend this literature by providing rare evidence on physician strikes; a distinct professional group whose core responsibilities in diagnosis, treatment formulation, and surgical interventions differ fundamentally from the continuous care and monitoring provided by nurses and midwives. These distinct clinical roles imply that physician disruptions shift healthcare delivery through entirely different structural mechanisms.

Our findings carry key implications for healthcare regulation and hospital contingency planning. Prolonged physician strikes generate severe threats to patient safety that cannot be resolved entirely by patient re-allocation or selective triaging. Because diminished care quality is responsible for half of the observed mortality spike, standard emergency coverage mandates during industrial actions may be insufficient for extended disruptions. Policymakers and hospital administrators must therefore establish more rigorous risk mitigation protocols before strikes occur, including strict minimum staffing thresholds for specialized care, coordinated regional patient diversion networks, and formal mechanisms to dynamically monitor and maintain clinical quality in non-striking fallback facilities.

2 Background

2.1 The German Hospital Sector

The German hospital sector is substantial, treating approximately 19 million patients annually across roughly 2,000 facilities housing 500,000 beds ([Federal Statistical Office, 2018](#)). Market ownership is divided among public, private, and non-profit organisations. Public facilities are further segmented by administrative level, encompassing municipal, federal state, national, and statutory accident insurance operators ([Goepfert and Conrad, 2013](#)). System financing follows a strict dual funding model: operating costs and clinical salaries are reimbursed by health insurance funds via the case-fixed German Diagnosis-Related Groups (G-DRG) system, whereas capital infrastructure investments are financed directly via the budgets of the 16 federal states ([Blümel et al., 2024](#)).

The focal point of our analysis rests on the nation’s 33 university hospitals. The majority of these institutions are operated directly by individual federal states and integrated into university medical faculties, embedding them in both clinical research and medical education. Crucially, all university hospitals hold “maximum care” status, serving as the sole regional hubs for highly specialized tertiary interventions that standard community clinics and local hospitals lack the capacity or equipment to perform.

University hospitals manage roughly 10% of Germany’s 19 million annual inpatients ([Federal Statistical Office, 2018](#)). Although these tertiary facilities constitute less than 2% of all hospitals, they disproportionately employ approximately 20% of the nation’s hospital physician workforce.³ On the demand side, institutional features of the German system grant patients unconstrained choice of healthcare providers; individuals can select any facility, and primary care physicians face no institutional mandates to refer patients to specific hospitals ([de Cruppé and Geraedts, 2017](#)). However, the system maintains a rigid structural separation between ambulatory and inpatient care, which legally prohibits hospital-based physicians

³Table A.1 in [Appendix A](#) illustrates the longitudinal evolution of physician and healthcare worker counts across all German hospitals and university hospitals from 2000 to 2010.

from treating outpatients ([Schreyögg et al., 2014](#)). Treatment costs are covered via mandatory universal health insurance spanning both statutory and private insurers ([Mossialos et al., 2016](#)).⁴

2.2 The 2006 University Hospital Strike in Germany

The 2006 German physician strike represents a watershed moment in European healthcare industrial relations, introducing a prolonged, supply-side shock to the nation's tertiary care sector. Historically, collective bargaining for all medical and non-medical public hospital employees was conducted under a centralized framework known as the Federal Employees Collective Agreement (*Bundes-Angestelltenarbeitsvertrag* or BAT). Under this centralized regime, the *Marburger Bund*, the professional association representing German physicians, acted as a specialized faction within a broader public sector bargaining coalition led by *ver.di*, the dominant service-sector trade union. However, this unified structure consistently compressed wage differentials and failed to address physician-specific grievances, such as routine 80-hour work weeks, uncompensated on-call duties, and declining relative wages. The tipping point occurred in late 2005 when public employers introduced a modernized framework, the Collective Agreement for the Public Service (*Tarifvertrag für den öffentlichen Dienst* or TVöD). Physicians argued that the TVöD would effectively implement net wage cuts and further devalue clinical overtime. Feeling marginalized by *ver.di*'s multi-profession bargaining goals, the *Marburger Bund* took the unprecedented step of breaking away from the joint coalition to establish itself as an autonomous labour union with independent strike capacity ([Greef, 2012](#)).

When subsequent independent negotiations with the Employers' Association of German Länder (*Tarifgemeinschaft der Länder* or TdL) collapsed, an unexpected wave of industrial action commenced in March 2006, lasting for 13 weeks until mid-June ([Marburger Bund Bundesverband, 2012](#)). The strike targeted state-run university hospitals, mobilizing up to

⁴Statutory health insurance covers the vast majority of the population (88%), while private insurance covers approximately 11% ([Bundesministerium für Gesundheit, 2019](#)). Although our administrative data includes both privately and publicly insured patients, it does not permit a distinction between these two categories.

22,000 physicians at its peak and causing severe operational disruptions across 24 of Germany's 33 academic medical centres. The roll-out of the strike followed a staggered timeline: 9 university hospitals initiated walkouts on the first day, while an additional 15 facilities joined sequentially over the following weeks (Martens, 2008). Crucially, the remaining nine university hospitals never participated in the dispute due to pre-existing, historical governance structures that legally insulated them from the national TdL bargaining unit, rendering their exemption plausibly exogenous to contemporaneous hospital performance.⁵ To prevent immediate danger to public life and satisfy constitutional mandates, the *Marburger Bund* and hospital administrators co-signed binding emergency care agreements, ensuring that emergency departments and obstetric units remained functional throughout the 90-day dispute. The strike concluded on June 16, 2006, with the signing of a landmark, stand-alone collective bargaining agreement for doctors (*Tarifvertrag für Ärztinnen und Ärzte* or TV-Ärzte), which secured wage increases of up to 18% and implemented strict caps on on-call shifts. Table 1 outlines the detailed chronological evolution of these events.

TABLE 1. Chronology of Events in the 2006 Hospital Strike

Date	Event
14 Apr 2005	<i>ver.di</i> and state employers launch centralized public sector bargaining
11 Sep 2005	Marburger Bund (<i>MB</i>) terminates collective bargaining partnership with <i>ver.di</i>
12 Oct 2005	<i>MB</i> and state employers (<i>TdL</i>) initiate autonomous contract negotiations
16 Mar 2006	Nationwide physician strike commences across state-run university hospitals
16 Jun 2006	<i>MB</i> and <i>TdL</i> sign stand-alone collective bargaining agreement (<i>TV-Ärzte</i>)

NOTE.— Based on Greef (2012) and Martens (2008). *MB* = Marburger Bund (physicians' trade union); *ver.di* = Vereinte Dienstleistungsgewerkschaft (general service sector union); *TdL* = Tariftgemeinschaft der Länder (employers' association of German federal states); *VKA* = Vereinigung der kommunalen Arbeitgeberverbände (association of municipal employers).

⁵Four hospitals (Berlin, Frankfurt, Gießen, and Marburg) did not participate because their respective federal states operated outside the central employers' association (TdL); Berlin left the association in 1994, and Hesse exited in 2004 (Greef, 2012). In Hamburg and Schleswig-Holstein (Kiel and Lübeck), university hospitals did not join the strike because they had previously transitioned into autonomous public corporations, placing them under independent regional bargaining units outside the TdL framework. The remaining clinics were excluded due to alternative governance structures: Mannheim operated under municipal collective bargaining due to its 100% municipal ownership, Bochum fell under federal insurance and corporate governance, and Wuppertal was a privately owned commercial facility operated by Helios.

3 Data

3.1 Data Sources and Administrative Registers

To evaluate the causal impact of the 2006 physician strike, we utilize patient-level administrative records from the German Hospital Statistics covering the years 2000 to 2008, provided by the Federal Statistical Office and the Statistical Offices of the Federal States ([FDZ der Statistischen Ämter des Bundes und der Länder, 2008](#)). This core dataset represents a complete census of all inpatient admissions across every hospital operating in Germany during the sample period. For each admission, the register provides granular clinical and demographic information, including the patient’s age, gender, regional place of residence, length of stay in days, a surgical procedure flag, and the diagnosis coded under the 3-digit International Classification of Diseases (ICD-10). Because German healthcare facilities are legally mandated to report these operational metrics, the administrative registry is complete ([Bundesamt für Justiz, 1990](#)).

To ensure that any documented changes in clinical outcomes are not selectively driven by changes in out-of-hospital mortality, such as high-risk patients being turned away and subsequently dying at home, we complement the hospital register with the official German Cause of Death Statistics. This database captures vital statistics on every registered death occurring within German territory. Using the individual’s district of residence and exact date of death, we merge the mortality data with our collected strike timeline records and collapse the data into district-date cells.⁶ Treatment assignment is based on the deceased’s district of residence at the time of death. Treated districts are defined as those hosting a striking university hospital, whereas control districts are those without a striking university hospital.

⁶The district (*Kreis*) is the intermediate administrative unit between the municipality (*Gemeinde*) and the federal state (*Bundesland*), holding explicit governance responsibilities over local healthcare infrastructure and public transport. In 2006, Germany comprised 439 districts.

3.2 Measuring Strike Participation and Timeline Reconstruction

Because there is no centralized administrative register tracking daily employee strike participation or industrial actions across German healthcare facilities ([Dribbusch, 2018](#)), we constructed a novel, institutional-level dataset of strike exposure. We conducted a systematic, manual audit of local, regional, and national newspaper archives published throughout 2006. We cross-referenced these media reports with official press releases from the *Marburger Bund* and historical archival records from individual hospitals to extract the exact calendar dates on which physicians at each facility initiated and terminated strike activity.

Our finalized strike database tracks all 33 German university hospitals, documenting that physicians across 24 facilities actively participated in the strike, whereas 9 facilities remained completely unaffected due to historical governance exemptions (see Section 2.2). Among the 24 treated institutions, the media audit reveals variation in treatment timing: a cohort of 9 university hospitals entered the dispute immediately on the first national strike day in March 2006, while the remaining 15 facilities sequentially initiated walkouts over the subsequent weeks.⁷

3.3 Outcome Variables and Sample Restrictions

For our primary hospital-level analysis, we collapse the underlying patient-level records to a balanced hospital-by-day cell structure. Our primary outcome variables are (i) the daily volume of inpatient admissions per hospital, capturing supply-side contractions and clinical rationing behaviour, and (ii) the daily in-hospital mortality rate at the facility level, operationalized as the number of patient deaths per 100 admissions on a given day. Following [Gruber and Kleiner \(2012\)](#) and [Geweke, Gowrisankaran and Town \(2003\)](#), we also construct an alternative mortality metric tracking patient deaths occurring within 10 days of admission. This restricted window mitigates potential bias stemming from strike-induced shifts in a patient’s overall length of stay. By capping the exposure window, we better isolate clinical mortality attributable to initial hospital admission care and rule out compounding factors

⁷See [Table A.2](#) in [Appendix A](#) for a detailed breakdown of strike dates for each hospital.

associated with post-strike facility transfers. For completeness, we also examine the absolute daily number of deaths per hospital.

To evaluate whether the labour disruption altered clinical prioritization and patient case-mix, we track shifts in demographic traits (age and gender) alongside objective proxies for clinical severity derived from detailed diagnosis codes. First, we construct an Ambulatory Care Sensitive (ACS) indicator based on the framework by [Sundmacher et al. \(2015\)](#) to identify hospitalizations that are theoretically avoidable through timely primary care intervention.⁸ Second, we build an objective emergency and urgency indicator utilizing the diagnosis-based machine learning classification developed by [Krämer, Schreyögg and Busse \(2019\)](#).⁹ Finally, we study operational changes by tracking the average length of stay (in days) and the baseline surgery rate (total surgeries per 100 admissions). Shifts in the length of stay capture capacity constraints, accelerated discharge protocols, or modified care intensity, while variations in surgical volume signal strategic rationing, such as the systematic postponement of elective procedures.

Our primary estimation sample restricts attention to acute inpatient admissions within the 33 university hospitals, tracking them daily from 2000 through 2008 to fully capture pre-trend dynamics, the active three-month strike period, and post-strike operational adjustments. We restrict the baseline panel exclusively to university hospitals to maintain a structurally balanced sample of maximum-care facilities and to avoid confounding variations from municipal-level hospital strikes that commenced in June 2006, for which reliable strike data is unavailable. The final aggregated sample consists of 143,744 hospital-day observations for our primary models and 80,241 district-date observations for our regional mortality analysis.

We operationalize baseline strike exposure using a binary indicator equal to one if a patient is admitted to a participating university hospital during its active strike window, and zero otherwise. Because this flow-based definition excludes individuals who were already

⁸The ACS metric identifies clinical conditions where effective ambulatory care typically precludes the necessity of an inpatient admission ([Sundmacher et al., 2015](#)).

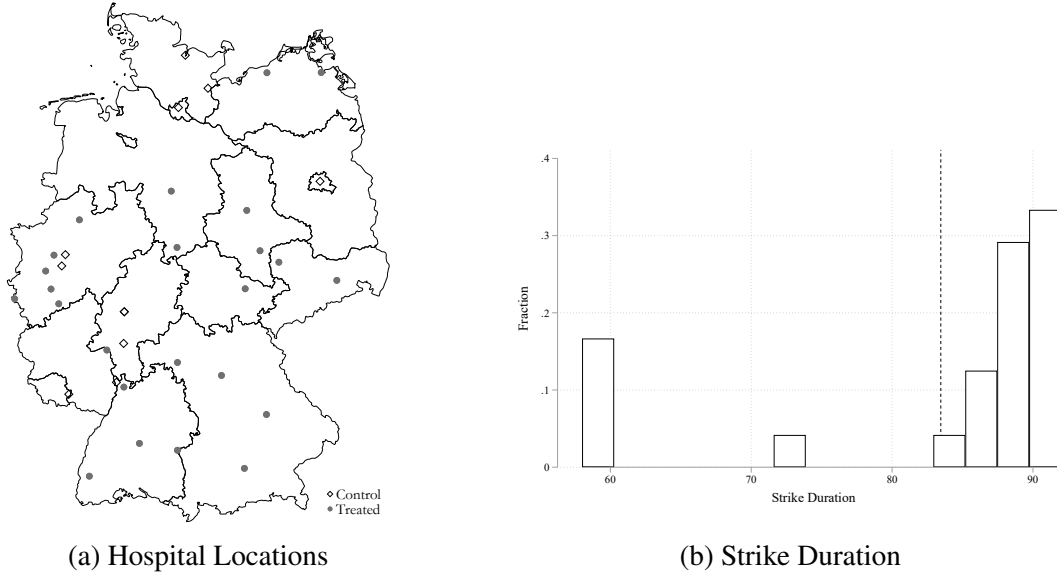
⁹[Krämer, Schreyögg and Busse \(2019\)](#) apply predictive algorithms across clinical codes to categorize admissions as elective or urgent. A hospitalization is empirically classified as an emergency if its predicted urgency index exceeds a threshold of 0.5.

hospitalized prior to the strike's onset, we also construct an alternative, stock-based treatment indicator as a robustness check. This alternative measure calculates the daily proportion of active inpatients whose hospital stays overlapped with a strike day at any point during their institutionalization (see Section 5.5). We favour our primary flow-based indicator for the baseline model because it is immune to mechanical bias arising from the strike's endogenous impact on a patient's length of stay.

3.4 Descriptive Sample Statistics and Covariate Balance

Figure 1 provides visual descriptive evidence mapping the spatial and temporal dimensions of the industrial dispute. Panel (a) illustrates the geographic distribution of our treatment and control groups across Germany, charting the locations of the 24 striking university hospitals alongside the 9 control facilities. As shown in the map, treatment and control hospitals are located in distinct administrative districts and generally span different federal states. This spatial separation mitigates concerns regarding direct, localized spillover effects contaminating our control group. Panel (b) details the distribution of strike durations among the 24 participating facilities. The mean strike duration was approximately 83 days, with a median of 88 days. This timeline represents a substantially more protracted disruption than those examined in prior healthcare literature; for comparison, Gruber and Kleiner (2012) report a mean nursing strike duration of 32 days and a median of 19 days. This marked difference underscores that our setting evaluates an extended supply-side shock rather than a brief institutional friction.

FIGURE 1. Geographic Distribution and Duration of Hospital Strikes



NOTE.— Own calculations based on collected strike data. Panel (a) illustrates the geographic distribution of our treatment cohort (the 24 strike-participating university hospitals) and control cohort (the 9 non-striking university hospitals) across German administrative districts. Panel (b) plots the empirical distribution of strike durations across all participating facilities. The vertical dashed line in Panel (b) demarcates the mean strike duration of approximately 83 days across all striking institutions.

Table 2 presents comprehensive baseline descriptive statistics for our primary outcomes and patient covariates, segmented by treatment status across the entire sample period. Because all 33 facilities share identical institutional mandates as maximum-care tertiary centres tied directly to university medical faculties, they exhibit strong covariate balance across both operational and demographic characteristics. This structural similarity suggests that non-striking university hospitals provide a valid counterfactual for the treated cohort, strengthening our identification strategy outlined in the next section.

TABLE 2. Baseline Descriptive Statistics by Treatment Status

Variable	Treated Cohort (N = 24)			Control Cohort (N = 9)		
	Mean	SD	Observations	Mean	SD	Observations
<i>Panel A. Hospital-Level data</i>						
Female share	0.49	0.05	84,986	0.49	0.07	58,758
Patient age (years)	47.38	3.89	84,986	48.99	5.28	58,758
Daily admissions (cases)	159.28	69.88	84,986	156.27	111.23	58,758
Log daily admissions	4.95	0.54	84,986	4.79	0.81	58,758
Length of stay (days)	8.12	1.59	84,986	8.26	2.13	58,758
Surgery rate (%)	37.06	22.17	84,986	33.57	22.03	58,758
Log daily inpatient deaths	0.83	0.59	68,985	0.92	0.66	38,410
In-hospital mortality rate (%)	1.67	1.45	84,986	1.94	2.17	58,758
10-day mortality rate (%)	0.91	1.11	84,986	1.08	1.64	58,758
Emergency admission rate (%)	27.65	12.18	84,648	30.03	13.22	58,508
Clinical urgency index	0.33	0.09	84,648	0.35	0.09	58,508
<i>Panel B. District-Level data</i>						
Female share among deceased	0.55	0.19	56,837	0.55	0.12	23,404
Age at death (years)	75.70	5.83	56,837	75.41	3.57	23,404
Population mortality rate (%)	0.26	0.10	56,837	0.28	0.07	23,404

NOTE.— Own calculations based on data from [FDZ der Statistischen Ämter des Bundes und der Länder \(2008\)](#). Baseline descriptive statistics segmented by the historical strike participation status of the university hospitals. Treated facilities comprise the 24 university hospitals that actively participated in the labour dispute, whereas control facilities consist of the 9 non-striking and legally exempt university hospitals. Hospital- and district-level metrics are weighted by the baseline average volume of pre-strike admissions per facility (panel A.) and population size (panel B.). Comprehensive variable definitions are detailed in [Table A.3](#) of [Appendix A](#).

4 Econometric Strategy

We employ several complementary econometric approaches to estimate the causal effects of the physician strike and systematically examine its underlying mechanisms. First, a hospital-level DiD design compares striking and non-striking university hospitals before and after the strike. Second, an analogous district-level DiD framework tracks population-level mortality to capture potential shifts in out-of-hospital mortality. Third, we integrate a double-selection LASSO algorithm ([Belloni, Chernozhukov and Hansen, 2014](#)) into our primary DiD specification to flexibly condition on a high-dimensional set of patient characteristics, isolating the mortality effect net of strike-induced case-mix alterations. Finally, we evaluate demand-side geographic spillovers by exploiting regional variation in pre-strike patient shares to test whether displaced individuals systematically shifted care toward nearby, non-striking

substitute hospitals.

Our identification strategy rests on the parallel trends assumption, which dictates that in the absence of the 2006 physician strike, the average trajectories of patient mortality, hospital admissions, and other clinical outcomes among the strike-participating university hospitals would have evolved in parallel with those of the non-striking control hospitals. We systematically evaluate the plausibility of this identifying assumption through two complementary approaches. First, we leverage a flexible event study framework to empirically test for the presence of any differential pre-treatment trends or anticipatory behaviour between the treated and control cohorts. Second, we perform formal balance tests on key time-invariant and baseline hospital characteristics to support the assumption that our control group provides a structurally valid counterfactual.¹⁰

4.1 Hospital-Level Analysis

We begin our empirical analysis by investigating the direct impacts of the physician strike on hospital admissions and in-hospital mortality. To this end, we contrast clinical outcomes before, during, and after the industrial dispute between university hospitals that experienced strike actions (the treatment group) and those that were structurally exempt from striking (the control group). Our baseline DiD regression specification is defined as follows:

$$Y_{it} = \alpha + \tau^H S_{it} + \mu_i + \nu_t + \Gamma_i \cdot t + \varepsilon_{it}, \quad (1)$$

where Y_{it} represents one of the facility-level operational or clinical outcomes detailed in Section 3.3 for hospital i on admission day t . The independent variable of interest, S_{it} , is a binary indicator equal to one if hospital i is participating in the strike on day t , and zero

¹⁰Table A.4 in Appendix A presents formal hospital-level balance tests using annual facility metrics (coding the entire year of 2006 as the treatment period). Regressing each characteristic on the treatment indicator alongside year fixed effects in Column (1) indicates that striking facilities are, on average, larger, with higher bed capacities and larger clinical staff sizes. Crucially, once we control for hospital fixed effects and facility-specific linear trends in Column (3), these scale differentials are entirely absorbed, and the two groups become well-balanced across all observable dimensions. Section 5.5 introduces an alternative control group as a robustness check.

otherwise.¹¹ The parameters μ_i denote hospital fixed effects, which control for unobserved, time-invariant institutional characteristics such as historical case volumes, overall bed capacity, and the structural share of medical specialists. The parameters v_t represent comprehensive date-of-admission fixed effects that absorb common non-linear temporal trends, including seasonal variations and lower baseline admission volumes on weekends. To fully capture these fluctuations, the date fixed effects include separate indicators for the calendar year, week of the year, and day of the week. The vector $\Gamma_i \cdot t$ incorporates hospital-specific linear time trends to allow for differential outcome trajectories across individual facilities over the sample window.¹² To account for institutional size heterogeneity and ensure our estimates reflect aggregate impacts, all regressions are weighted by each hospital’s baseline average daily admission volume. Finally, idiosyncratic errors are represented by ε_{it} , with standard errors clustered at the hospital level to adjust for arbitrary serial correlation within facilities over time.

Our estimation strategy exploits variation in strike exposure across space and time, comparing the trajectory of outcomes at striking hospitals to that of never-striking hospitals before and after the strike. Because non-participating hospitals were structurally exempt from the national bargaining unit for historical reasons unrelated to contemporaneous performance, the comparison group is plausibly exogenous. The key parameter of interest, τ^H , therefore identifies the average treatment effect on the treated (ATT); the causal impact of strike participation on healthcare delivery and patient survival at affected university hospitals.

To visually validate our underlying parallel trends assumption and trace the dynamic trajectory of the strike’s impact over time, we extend our baseline DiD framework into an event study specification. This flexible design allows us to empirically test for pre-treatment anticipatory effects and evaluate the persistence of operational shocks during and after the

¹¹That is, $S_{it} = \mathbb{1}(i \in \mathcal{H}^S) \times \mathbb{1}(t \in \mathcal{T}^S)$ where $\mathbb{1}(\cdot)$ denotes the indicator function. The subset $\mathcal{H}^S \in \mathcal{H}$ represents the group of university hospitals that actively participated in the industrial action, and the subset $\mathcal{T}^S \in \mathcal{T}$ spans the calendar days of strike participation.

¹²For transparency, we also report estimates from a more parsimonious specification that omits hospital-specific trends.

industrial dispute. We estimate the following event study model:

$$Y_{it} = \alpha + \sum_{\substack{k=-K \\ k \neq -1}}^K \tau_k^H (S_i \times \mathbb{1}(t \in k)) + \mu_i + \nu_t + \Gamma_i \cdot t + \varepsilon_{it}, \quad (2)$$

where the term $\mathbb{1}(t \in k)$ represents an indicator function equal to one if admission day t falls within the specific bi-monthly relative event calendar block k , where $k = -1$ serves as the omitted baseline reference period (corresponding to January/February 2006). We aggregate the temporal indicators into bi-monthly intervals to balance statistical precision with descriptive insights on treatment effect heterogeneity. The parameter K denotes the maximum number of leads and lags included in our estimation window, capturing the full pre-strike baseline year of 2005, the active strike period, and the post-dispute recovery phase through 2007.

The parameters of interest are the vector of coefficients τ_k^H , which map out the dynamic treatment effects. The pre-treatment lead coefficients provide a direct test of the parallel trends assumption. The active strike coefficients capture the net dynamic impact of the healthcare supply shock during the strike period. Finally, the post-strike coefficients reveal whether the regional healthcare system experienced any post-dispute anomalies or returned to baseline equilibrium after the strike. To remain consistent with our main specification, the regressions are weighted by baseline average daily admissions per hospital, and standard errors are clustered at the hospital level.¹³

4.2 District-Level Analysis

While the facility-level analysis isolates the direct impacts of the strike within the affected institutions, it may yield an incomplete picture of the overall public health consequences if

¹³As a robustness check, we account for differences in the exact timing of strike onset across hospitals. While our baseline specification groups time into bi-monthly bins, strike start dates differ by up to one month, so hospitals may be exposed to treatment for different fractions of the first treatment period. We therefore re-estimate specification (2) at the monthly level using the interaction-weighted estimator of [Sun and Abraham \(2021\)](#), with never-treated hospitals as the comparison group. The resulting estimates closely mirror our baseline findings.

patients were systematically displaced across space. For instance, if high-risk individuals were turned away from striking university hospitals and subsequently died at home or in transit, relying solely on inpatient records would artificially underestimate the true lethal cost of the strike. Conversely, if displaced patients successfully substituted university hospital care with alternative local treatments without facing adverse health outcomes, the net population-level impact might be negligible.

To capture these broader spatial dynamics, visually validate our underlying parallel trends assumption at the macro level, and trace the dynamic trajectory of the strike’s impact on regional population health, we implement an analogous DiD event study framework at the regional district level. We map the universe of individual deaths from the official German Cause of Death Statistics to the deceased’s administrative district of residence and estimate the following district-level panel regression model:

$$M_{dt} = \alpha + \sum_{\substack{k=-K \\ k \neq -1}}^K \tau_k^D (S_d \times \mathbb{1}(t \in k)) + \mu_d + \delta_t + \beta X_{dt} + \varepsilon_{dt}, \quad (3)$$

where M_{dt} tracks the population-adjusted mortality rate (expressed as deaths per 10,000 inhabitants) in district d on day t . The variable S_d is a cross-sectional treatment indicator equal to one if administrative district d hosts a strike-participating university hospital, and zero otherwise. The indicator function $\mathbb{1}(t \in k)$ maps daily panel observations into their respective bi-monthly event bins k . The model includes district fixed effects μ_d , which control for time-invariant regional health and demographic characteristics, and date fixed effects δ_t , which absorb common non-linear time trends. The vector X_{dt} incorporates controls for time-varying district-level characteristics, such as the share of females in the regional population. Regressions are weighted by the baseline district population, and standard errors are clustered at the district level.

Districts without any university hospital are excluded from this baseline regional specification to maintain a tightly matched counterfactual sample of academic medical catchment areas. Furthermore, we estimate the effects of the strike on district mortality only until May/June

2006, as another strike occurred at the district level thereafter and data for this subsequent strike are not available. The key parameters of interest are the τ_k^D coefficients, which measure the impact of the strike on district mortality relative to the two-month bin before the strike.

4.3 Double-Selection LASSO and Case-Mix Decompositions

It is important to recognize that the physician strike likely induced systemic shifts in inpatient composition. While emergency care remained operational under binding agreements, elective admissions and deferrable surgical procedures were substantially rationed. Because emergency or high-acuity patients exhibit systematically worse health outcomes and higher baseline risk than elective cohorts, such changes in the underlying case-mix will mechanically confound our baseline DiD estimates. Consequently, while the raw baseline coefficient captures the aggregate causal effect of the strike period on in-hospital mortality, it conflates true quality-of-care degradation with strategic patient triaging, failing to isolate the primary parameter of interest for healthcare policy.

To determine whether clinical quality of care was genuinely compromised, our empirical estimation must condition on a structurally comparable patient population. We rigorously address this selection concern by implementing the double-selection LASSO covariate selection framework introduced by [Belloni, Chernozhukov and Hansen \(2014\)](#). To capture strike-induced composition shifts across high-dimensional clinical profiles, we leverage the extensive array of patient and institutional characteristics available in our administrative registries. Specifically, we compile individual indicators for age, gender, detailed 3-digit ICD-10 diagnosis codes, and objective indices of clinical urgency and deferability, yielding a large dictionary of 337 potential covariates.

Relying on standard linear controls across this high-dimensional dictionary introduces severe risks of overfitting, multi-collinearity, and arbitrary functional form misspecification. The double-selection LASSO algorithm resolves these challenges through a principled, two-stage variable selection process. In the first stage, a treatment prediction LASSO regression model uses the full dictionary of 337 patient traits to predict strike exposure (i.e., admission to

a participating facility during the active strike window), systematically identifying covariates that capture strike-induced patient selection and institutional triaging strategies. In the second stage, an outcome prediction LASSO regression model uses the same dictionary to predict the primary clinical outcome, isolating covariates that serve as objective predictors of patient mortality.¹⁴

Ultimately, we include the union of all variables selected across both stages as linear controls in our primary facility-level specification:

$$Y_{it} = \alpha + \tau^{H_{adj}} S_{it} + \mu_i + \nu_t + \Gamma_i \cdot t + \mathbf{X}_{it}' \Delta + \varepsilon_{it}, \quad (4)$$

where all baseline parameters and fixed effects are defined as before, and $\mathbf{X}_{it}$ represents the vector of selected case-mix covariates aggregated to the hospital-day cell structure. The key parameter of interest, $\tau^{H_{adj}}$, identifies the causal impact of the strike on inpatient mortality net of all strike-induced changes in observed patient composition. This case-mix-adjusted estimate isolates the quality-of-care channel, delivering a policy-relevant parameter evaluating healthcare supply resilience.

4.4 Demand-Side Spillover Analysis

When admissions at striking hospitals contract and clinical prioritization shifts toward higher-severity cases, healthier patients may respond by delaying hospitalization, seeking care at alternative facilities, or substituting inpatient care with ambulatory services. Our subsequent evaluation of post-strike admissions and catch-up mortality patterns will offer direct evidence on whether treatment postponement serves as a widespread behavioural response. To capture the potential geographic displacement of patients, we exploit regional variation in pre-strike reliance on the affected tertiary hospitals.

Using the pre-strike year of 2005 as a benchmark, we compute for each administrative district the baseline share of patients who received care at any strike-participating university

¹⁴To avoid over-fitting, we impose a penalty on the number of predictors, where the tuning parameter has been chosen by five-fold cross-validation.

hospital. This baseline continuous share serves as a proxy for localized treatment intensity: the operational capacity and volume pressures exerted on neighbouring, non-treated hospitals should be highest in districts that historically depended most heavily on striking facilities. We integrate this pre-strike patient share into the following district-level regression model:

$$Y_{dt} = \alpha + \tau^{D_{\text{spillover}}} SS_{dt} + \mu_d + \nu_t + \Gamma_d \cdot t + \varepsilon_{dt}, \quad (5)$$

where Y_{dt} denotes a specific aggregated non-university hospital outcome for district d on admission day t . The parameters μ_d and ν_t represent district and date-of-admission fixed effects, respectively, while $\Gamma_d \cdot t$ incorporates district-specific linear time trends. The coefficient $\tau^{D_{\text{spillover}}}$ identifies the causal effect of demand-side displacement on neighbouring, non-striking hospitals by capturing how a one-unit increase in a district's baseline reliance on striking facilities impacts local clinical volumes and patient mortality during the strike period. The primary independent variable of interest, SS_{dt} , is a continuous treatment intensity metric that interacts the active strike period with a district's baseline geographic reliance on striking facilities, operationalized as:

$$SS_{dt} = \mathbb{1}(t \in \mathcal{T}^S) \times \left(\frac{\sum_{i \in \mathcal{H}^S} \text{Admissions}_{id,2005}}{\sum_{i \in \mathcal{H}} \text{Admissions}_{id,2005}} \right), \quad (6)$$

where $\mathbb{1}(t \in \mathcal{T}^S)$ is an indicator function evaluating to one if admission day t falls within the active strike window in 2006, and zero otherwise. The second term is a continuous weight representing the baseline geographic reliance of district d on striking facilities during the pre-strike benchmark year of 2005. Specifically, the numerator sums all inpatient admissions originating from district d that were treated across the set of strike-participating university hospitals ($\mathcal{H}^S$), while the denominator captures the total volume of hospital admissions from district d across the universe of all domestic clinics ($\mathcal{H}$). Standard errors are clustered at the district level, and regressions are weighted by baseline district population.

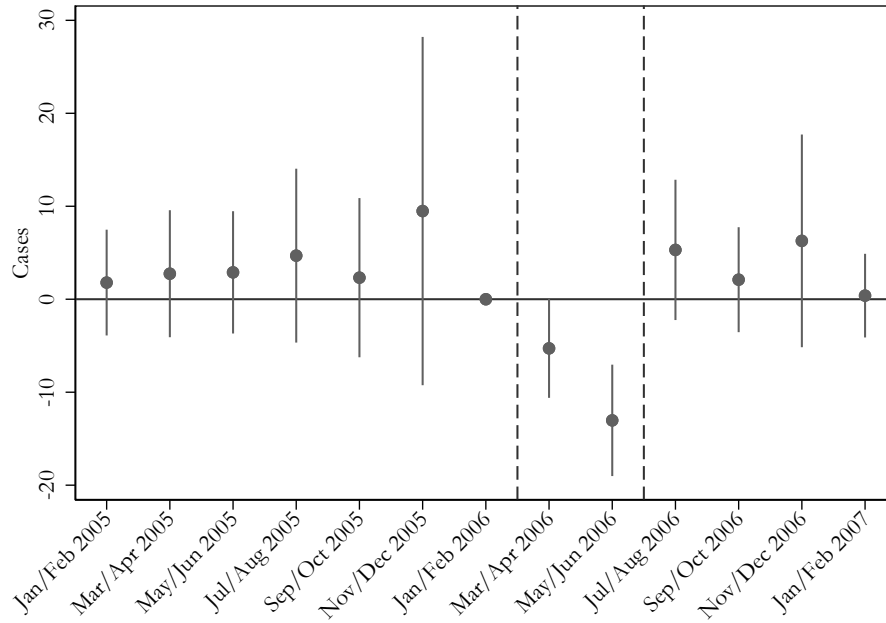
5 Results

5.1 Impact on Hospital Admissions and Mortality

We first investigate whether the 2006 physician strike affected the volume of daily hospital admissions. [Figure 2](#) illustrates the bi-monthly event study estimates for the relative change in hospital admissions across striking and non-striking university hospitals, estimated via Equation (2). Reassuringly, the point estimates in the pre-treatment periods leading up to the dispute are tightly bounded and statistically indistinguishable from zero. This absence of any pre-existing divergent trend strongly suggests that the parallel trends assumption required for a causal interpretation of our DiD estimator is satisfied. Turning to the estimates for the active strike period itself, demarcated by the vertical dashed lines, we document a sharp, statistically significant relative contraction in admissions at strike-participating hospitals. This reduction is most pronounced during May and June 2006, coinciding with the strike’s peak mobilization phase. Following the formal resolution of the dispute, the dynamic coefficient estimates immediately revert to statistical insignificance, displaying no evidence of a post-strike volume rebound. This lack of an operational “catch-up” effect suggests that the labour disruption caused a net overall deficit in total hospital admissions rather than a mere intertemporal reallocation of elective treatments.¹⁵

¹⁵Panels A and B of [Figure A.1](#) in [Appendix A](#) confirms this interpretation, demonstrating that the divergence is entirely driven by a sharp contraction in absolute admissions within treated hospitals rather than any anomalous operational shifts within the control group.

FIGURE 2. Event Study Results: Hospital Admissions



NOTE.— Own calculations based on data from [FDZ der Statistischen Ämter des Bundes und der Länder \(2008\)](#). The figure plots event study coefficient estimates for daily hospital admissions. All underlying specifications condition on hospital fixed effects, day-of-the-week fixed effects, week-of-the-year fixed effects, and calendar year fixed effects. The vertical dashed lines demarcate the initiation and formal resolution of the strike period. Shaded areas (or error bars) represent 95% confidence intervals computed from robust standard errors clustered at the hospital level. Regressions are weighted by the baseline average volume of pre-strike admissions per hospital. Comprehensive variable definitions are detailed in [Table A.3](#) of [Appendix A](#).

Formal DiD estimates are reported in [Table 3](#), confirming the structural patterns uncovered in the event study. Specifically, when conditioning on a comprehensive set of temporal indicators, hospital fixed effects, and facility-specific linear time trends, the point estimates in Columns (1) and (2) indicate a statistically significant decrease of 19 to 21 daily admissions on average, representing a 12% reduction relative to the baseline mean.

TABLE 3. Difference-in-Differences Estimates: Hospital Admissions

Dependent Variable	Observations	Baseline Mean	(1)	(2)
Daily Admissions	143,744	158.32	−21.460*** (4.646)	−19.041*** (2.596)
Log Daily Admissions	143,744	4.90	−0.122*** (0.019)	−0.118*** (0.014)
Calendar Year Fixed Effects			✓	✓
Hospital Fixed Effects			✓	✓
Hospital Linear Trends				✓

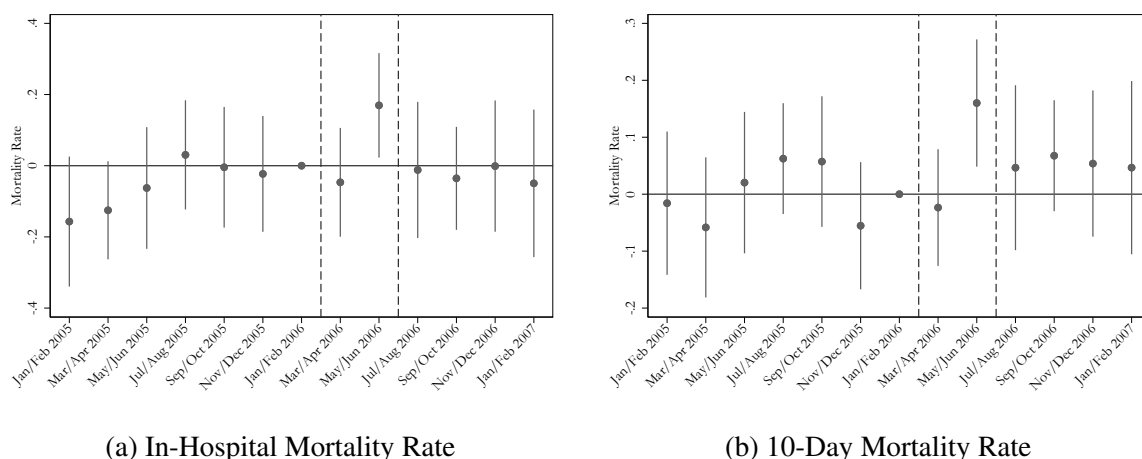
NOTE.— Own calculations based on data from [FDZ der Statistischen Ämter des Bundes und der Länder \(2008\)](#). Robust standard errors clustered at the hospital level are reported in parentheses. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. All models include hospital fixed effects, day-of-the-week fixed effects, week-of-the-year fixed effects, calendar year fixed effects, and hospital-specific linear time trends. Regressions are weighted by the baseline average volume of pre-strike admissions per hospital. Comprehensive variable definitions are detailed in [Table A.3](#) of [Appendix A](#).

Next, we analyse the consequences of this supply contraction on inpatient health outcomes. [Figure 3](#) presents the corresponding event study trajectories for both the in-hospital mortality rate and the 10-day mortality rate. Utilizing a 10-day post-admission window mitigates potential econometric bias stemming from strike-induced shifts in a patient’s overall length of stay, thereby isolating deaths more directly attributable to the initial phase of clinical care while filtering out compounding factors from subsequent events. Mirroring the admissions framework, the pre-treatment leads for both mortality metrics reveal identical trends between the treatment and control cohorts prior to March 2006, further substantiating our empirical approach.

During the intensive strike months of May and June, however, we document a stark reversal in these trajectories, characterized by a statistically significant relative surge in both raw in-hospital and 10-day mortality rates within treated facilities. These elevated risk levels revert back to baseline immediately upon the strike’s cessation.¹⁶

¹⁶Panels C and D of [Figure A.1](#) in [Appendix A](#) verifies that this mortality spike is mainly driven by an increase in mortality rates among treated hospitals.

FIGURE 3. Event Study Estimates: In-Hospital Mortality



NOTE.— Own calculations based on data from [FDZ der Statistischen Ämter des Bundes und der Länder \(2008\)](#). The panels plot event study coefficient estimates for the global in-hospital mortality rate (Panel a) and the restricted 10-day in-hospital mortality rate (Panel b). All underlying regression specifications condition on hospital fixed effects, day-of-the-week fixed effects, week-of-the-year fixed effects, and calendar year fixed effects. The vertical dashed lines demarcate the initiation and formal resolution of the strike period. Shaded areas (or error bars) represent 95% confidence intervals computed from robust standard errors clustered at the hospital level. Regressions are weighted by the baseline average volume of pre-strike admissions per hospital. Comprehensive variable definitions are detailed in [Table A.3](#) of [Appendix A](#).

The formal DiD coefficients corresponding to these dynamic paths are presented in Columns (1) and (2) of [Table 4](#). The point estimates imply that the daily in-hospital mortality rate increased by approximately 0.16 percentage points during the strike period. This represents a substantial 9% surge relative to the pre-strike baseline mean of 1.75 deaths per 100 admissions. The corresponding estimates for the 10-day mortality rate are somewhat lower at 0.07 percentage points (a 7.3% increase), yet remain statistically significant at the 5% level. Finally, the estimates reported in the third row of the table indicate that the absolute daily number of inpatient deaths dropped by 0.07 cases from a baseline mean of 2.46; a statistically insignificant reduction of 2.8%. This drop is mathematically consistent with the other results on admissions and the mortality rate. Rather than indicating a surge in the absolute volume of fatalities, these joint patterns demonstrate that the strike triggered a compositional shift in the underlying patient pool. The documented increase in the mortality rate thus reflects a higher concentration of clinical risk among a smaller, systematically frailer baseline of patients.

TABLE 4. Difference-in-Differences Estimates: In-Hospital Mortality and Case-Mix Adjustments

Dependent Variable	Observations	Mean	Baseline Specification		Covariate Adjusted	
			(1)	(2)	(3)	(4)
In-Hospital Mortality Rate	143,744	1.75	0.159*** (0.043)	0.165*** (0.041)	0.069* (0.037)	0.072** (0.035)
10-Day Mortality Rate	143,744	0.96	0.074** (0.036)	0.073** (0.035)	0.012 (0.030)	0.011 (0.029)
Daily Number of Deaths	143,744	2.46	−0.099 (0.087)	−0.069 (0.056)	−0.093 (0.076)	−0.068 (0.054)
Predicted Mortality Rate	143,744	1.81	0.000 (0.000)	0.000 (0.000)		
Covariate Adjustment					✓	✓
Calendar Year Fixed Effects			✓	✓	✓	✓
Hospital Fixed Effects			✓	✓	✓	✓
Hospital Linear Trends				✓		✓

NOTE.— Own calculations based on data from [FDZ der Statistischen Ämter des Bundes und der Länder \(2008\)](#). Robust standard errors clustered at the hospital level are reported in parentheses. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. All models include hospital fixed effects, day-of-the-week fixed effects, week-of-the-year fixed effects, calendar year fixed effects, and hospital-specific linear time trends. Regressions are weighted by the baseline average volume of pre-strike admissions per hospital. Comprehensive variable definitions are detailed in [Table A.3](#) of [Appendix A](#).

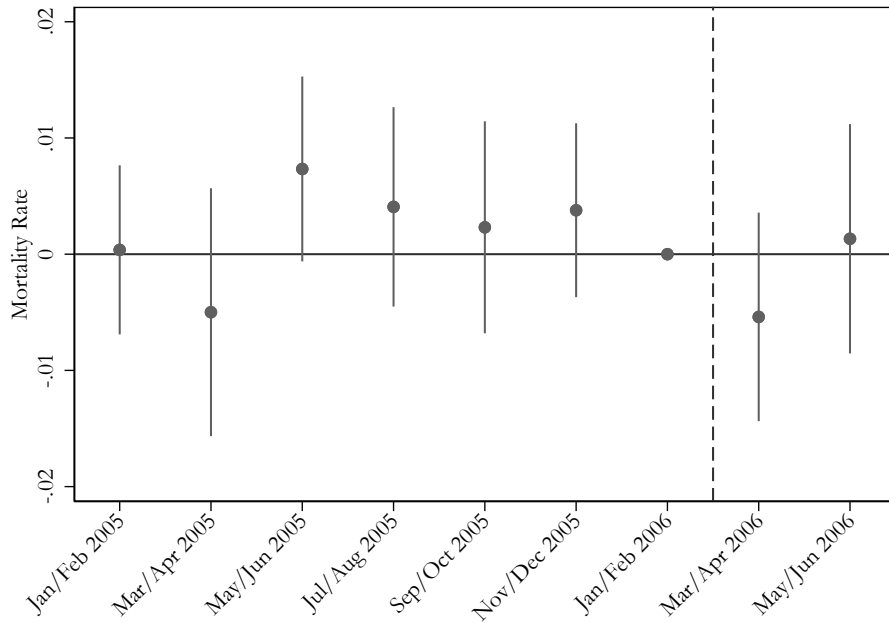
5.2 Impact on Out-of-Hospital Mortality

To evaluate the broader systemic costs of the strike, we test whether the hospital-level patterns generated a detectable footprint on population health at the regional level. Reconstructing vital statistics at the district level is essential for capturing potential out-of-hospital mortality spillovers, which could emerge if high-risk patients were prematurely discharged or failed to secure timely admission for acute, life-threatening conditions. However, because university hospitals accounted for only 2.3% of all national fatalities in the year prior to the dispute, any population-wide mortality footprint is likely to be small.

[Figure 4](#) presents the district-level event study estimates comparing population-adjusted out-of-hospital mortality rates between treated and control districts, as specified in Equation (3). Treated districts are defined as those hosting a strike-participating university hospital, while control districts comprise districts hosting non-striking facilities. The estimated co-

efficients remain tightly bounded around zero and are statistically insignificant throughout the entire active strike window. This lack of a systematic trend demonstrates that the severe supply-side shocks and localized clinical risks observed within striking institutions did not spill over to trigger a broader public health crisis outside the hospital setting.

FIGURE 4. Event Study Estimates: District Mortality



NOTE.— Own calculations based on data from [FDZ der Statistischen Ämter des Bundes und der Länder \(2008\)](#). The figure plots event study coefficient estimates for the population-adjusted district mortality rate (expressed as deaths per 10,000 residents). All underlying regression specifications condition on district fixed effects, day-of-the-week fixed effects, week-of-the-year fixed effects, and calendar year fixed effects. The vertical dashed lines demarcate the initiation and formal resolution of the strike period. Shaded areas (or error bars) represent 95% confidence intervals computed from robust standard errors clustered at the district level. Regressions are weighted by baseline district population. Comprehensive variable definitions are detailed in [Table A.3](#) of [Appendix A](#).

5.3 Disentangling Quality of Care and Patient Composition Shocks

While our baseline DiD estimates identify the aggregate causal impact of the strike period, the resulting coefficient conflates genuine quality-of-care degradation with selective case-mix adjustments. To isolate the direct effect of the labour disruption on in-hospital mortality from strike-induced shifts in the underlying patient pool, we implement the double-selection

LASSO covariate adjustment framework outlined in Section 4.3. Columns (3) and (4) of Table 4 report these conditional estimates. Once we rigorously control for the high-dimensional dictionary of patient characteristics and clinical codes, the strike’s estimated impact on the in-hospital mortality rate is attenuated by approximately half, falling to a moderate but statistically significant 0.07 percentage points (a 4.1% increase over the baseline mean). This structural attenuation confirms that the raw mortality surge is driven by two distinct, concurrent channels: half of the effect stems from strategic triaging, where patients in poorer baseline health were systematically prioritized for admission, while the remaining half reflects a localized reduction in clinical care quality due to physician supply constraints.

Although the double-selection LASSO procedure isolates a cleaner parameter net of selection bias, the shrinkage mechanism does not yield structural coefficients for individual control variables, leaving the specific dimensions of the case-mix shock unobserved. To directly uncover how patient characteristics and hospital rationing behaviours adjusted during the 90-day dispute, we estimate our primary DiD specification across a series of explicit patient-level traits and institutional outcomes. Table 5 reports these results, tracking shifts in demographic composition, clinical acuity indicators, average length of stay, and surgical volumes. While attributes such as age and gender reflect immutable pre-admission patient characteristics, operational metrics like length of stay and surgical rates capture dynamic hospital coping strategies and capacity constraints.

The point estimates in Table 5 demonstrate that the strike triggered significant composition shocks and operational reallocations. First, the average age of admitted patients significantly decreased by 0.41 years. This downward shift is entirely consistent with selective triaging toward high-acuity cases, as emergency admissions are systematically younger on average than the elective cohorts undergoing chronic or age-related procedures.¹⁷ Second, in line with this demographic shift, the emergency admission rate and the clinical urgency index experienced sharp, statistically significant increases. Third, we document a significant expansion in the average length of stay by 0.24 days, suggesting that the remaining inpatient cohort consisted

¹⁷Figure A.2 in Appendix A illustrates this underlying age distribution, contrasting emergency and non-emergency inpatient populations.

of systematically sicker individuals requiring more resource-intensive care. Conversely, we detect no statistically significant shifts in gender composition or the rate of Ambulatory Care Sensitive (ACS) conditions. Finally, the baseline surgery rate exhibits a substantial, albeit marginally significant, reduction of approximately 4.5 percent. This sharp contraction may reflect the systematic postponement of elective, deferrable surgical procedures as university hospitals rationed their depleted physician workforces to maintain core emergency services. Taken together, these results suggest that the remaining inpatient cohort during the strike faced a higher concentration of baseline clinical risk.

TABLE 5. Difference-in-Differences Estimates: Patient Composition and Hospital Decisions

Dependent Variable	Observations	Baseline Mean	(1)	(2)
<i>Panel A. Patient Composition Indicators</i>				
Patient Age (years)	143,744	47.89	−0.369** (0.157)	−0.411*** (0.146)
Female Share	143,744	0.49	0.002 (0.002)	0.002 (0.002)
ACS Admission Rate	143,744	20.30	−0.188 (0.221)	−0.301 (0.183)
Emergency Admission Rate	143,156	28.41	1.558*** (0.349)	1.505*** (0.348)
Clinical Urgency Index	143,156	0.34	0.012*** (0.003)	0.012*** (0.002)
Deferrability Index	143,549	0.12	0.003*** (0.001)	0.003*** (0.001)
<i>Panel B. Institutional Hospital Decisions</i>				
Length of Stay (days)	143,744	8.17	0.255*** (0.055)	0.243*** (0.052)
Surgery Rate	143,744	35.95	0.016 (1.409)	−1.629* (0.930)
Calendar Year Fixed Effects			✓	✓
Hospital Fixed Effects			✓	✓
Hospital Linear Trends				✓

NOTE.— Own calculations based on data from [FDZ der Statistischen Ämter des Bundes und der Länder \(2008\)](#). Robust standard errors clustered at the hospital level are reported in parentheses. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. All models include hospital fixed effects, day-of-the-week fixed effects, week-of-the-year fixed effects, calendar year fixed effects, and hospital-specific linear time trends. Regressions are weighted by the baseline average volume of pre-strike admissions per hospital. Comprehensive variable definitions are detailed in [Table A.3 of Appendix A](#).

To further understand how the strike reshaped clinical risk and operations, we partition our

sample by baseline emergency status. This modification allows us to test whether the strike's consequences were concentrated among non-emergency cohorts, who could plausibly defer treatment, foregone care, or seek alternative facilities, or whether the disruption breached the institutional safeguards designed to protect acute emergency admissions requiring immediate clinical intervention.

Table 6 presents separate DiD estimates for emergency and non-emergency patient subsamples. Consistent with strategic supply-side rationing, the contraction in daily admissions is more than twice as large for non-emergency cases (15.5%) than for emergency admissions (7.1%), confirming a substantial relative shift in the inpatient pool toward higher-acuity cases. The mortality effects are larger and exclusively statistically significant within the non-emergency cohort. This asymmetry holds in both absolute and relative terms, despite the fact that the baseline mortality rate for emergency patients is more than double that of non-emergency cases.

Taken together, the results in Table 6 demonstrate that while formal emergency care agreements successfully shielded acute emergency patients from elevated mortality, the systemic disruptions heavily penalized the remaining non-emergency inpatient population. Because the sharp decline in elective admissions left a systematically frailer, higher-risk pool of non-emergency patients hospitalized on understaffed regular wards, their localized mortality risks escalated significantly. This mechanism is further substantiated by a substantially larger strike-induced expansion in the length of stay for non-emergencies.¹⁸

¹⁸In addition, our conclusions are supported by the complementary heterogeneity analysis across quartiles of the deferability index presented in Table A.5 of Appendix A. The clinical deferability index is constructed following the frameworks of Card, Dobkin and Maestas (2009) and Halla, Liu and Liu (2019).

TABLE 6. Difference-in-Differences Estimates: Heterogeneity Analysis by Admission Urgency

Dependent Variable	Emergency Admissions			Non-Emergency Admissions		
	Mean	(1)	(2)	Mean	(1)	(2)
Daily Admissions	143.06	−18.516*** (4.197)	−16.631*** (2.294)	164.18	−21.976*** (4.640)	−19.793*** (2.700)
Log Daily Admissions	3.48	−0.069*** (0.016)	−0.071*** (0.010)	4.57	−0.159*** (0.023)	−0.155*** (0.020)
Mortality Rate	3.37	0.133 (0.102)	0.143 (0.098)	1.27	0.207*** (0.044)	0.207*** (0.043)
10-Day Mortality Rate	2.21	0.071 (0.080)	0.072 (0.079)	0.55	0.080*** (0.026)	0.074*** (0.025)
Daily Number of Deaths	1.19	−0.064 (0.049)	−0.055 (0.038)	1.30	−0.029 (0.046)	−0.015 (0.033)
Length of Stay (days)	9.31	0.241** (0.107)	0.232** (0.092)	8.00	0.368*** (0.061)	0.358*** (0.057)
Surgery Rate	24.10	0.047 (1.220)	−1.256 (0.786)	40.31	0.074 (1.591)	−1.798* (1.063)
Calendar Year Fixed Effects		✓	✓		✓	✓
Hospital Fixed Effects		✓	✓		✓	✓
Hospital Linear Trends			✓			✓

NOTE.— Own calculations based on data from [FDZ der Statistischen Ämter des Bundes und der Länder \(2008\)](#). Robust standard errors clustered at the hospital level are reported in parentheses. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. All models include hospital fixed effects, day-of-the-week fixed effects, week-of-the-year fixed effects, calendar year fixed effects, and hospital-specific linear time trends. Regressions are weighted by the baseline average volume of pre-strike admissions per hospital. Comprehensive variable definitions are detailed in [Table A.3](#) of [Appendix A](#).

5.4 Demand-Side Spillovers to Neighbouring Hospitals

To evaluate demand-side geographic displacement, we estimate our district-level spillover model specified in Equation (5) across a series of non-university hospital outcomes. [Table 7](#) reports the estimated coefficient of interest, $\tau^{\text{spillover}}$, which captures the continuous treatment intensity of regional patient diversion based on pre-strike referral shares. The point estimates indicate that non-striking local hospitals proximate to treated academic medical centres experienced a statistically significant increase in baseline admission volumes during the physician strike. Specifically, a fallback hospital located in a district where 10% of hospitalizations took place in a striking university hospital prior to the dispute experienced an average increase

of 0.8% in admissions during the active strike window.¹⁹ Reassuringly, this demand shock did not trigger capacity constraints or degrade the quality of care within fallback institutions; the corresponding parameter estimates for both regional mortality rates and average length of stay are precisely estimated and statistically indistinguishable from zero at conventional significance levels.

TABLE 7. Difference-in-Differences Estimates: Demand-Side Spillovers

Dependent Variable	Observations	Baseline Mean	(1)	(2)
Log Daily Admissions	1,041,569	4.25	0.102*** (0.031)	0.078*** (0.020)
Mortality Rate (%)	1,041,569	2.57	-0.118 (0.124)	0.039 (0.094)
10-Day Mortality Rate (%)	1,041,569	1.62	-0.070 (0.090)	-0.014 (0.081)
Length of Stay (days)	1,041,736	8.19	0.483*** (0.143)	0.105 (0.109)
Calendar Year Fixed Effects			✓	✓
District Fixed Effects			✓	✓
District Linear Trends				✓

NOTE.— Own calculations based on data from [FDZ der Statistischen Ämter des Bundes und der Länder \(2008\)](#). Robust standard errors clustered at the district level are reported in parentheses. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. All models include district fixed effects, day-of-the-week fixed effects, week-of-the-year fixed effects, calendar year fixed effects, and district-specific linear time trends. Regressions are weighted by the baseline average volume of pre-strike admissions per hospital. Comprehensive variable definitions are detailed in [Table A.3](#) of [Appendix A](#).

5.5 Robustness and Sensitivity Analysis

We conduct several rigorous sensitivity checks to verify the stability of our primary findings which are all reported in [Appendix A](#). First, estimating hospital-level placebo regressions by backdating a fictitious strike to the identical calendar window in 2004 yields small, statistically insignificant coefficients, ruling out seasonal confounding ([Table A.6](#)). Second, our results remain highly robust to utilizing an alternative control group comprising all hospitals in

¹⁹Crucially, these spillover dynamics are captured exclusively within non-university facilities. Because non-university hospitals are omitted from our primary estimation sample, our primary university control group remains uncontaminated by these localized demand shifts.

Hamburg and Berlin (Table A.7)²⁰, as well as to operationalising a stock-based treatment indicator tracking the daily proportion of active inpatients whose stays overlapped with a strike day (Table A.8). Third, formal pre-trend testing reveals that the pre-period coefficients are tightly bounded around zero (Table A.9); applying the Roth et al. (2022) framework confirms that even a conservative linear extrapolation of these minor pre-trends cannot account for the magnitude of our estimated treatment effects.

Finally, we assess whether differences in the exact timing of strike onset affect our results. While our baseline specification aggregates daily data into bi-monthly intervals, the true strike start dates differed across treated academic medical centres by up to four weeks. Consequently, hospitals are not necessarily exposed to treatment for the identical fraction of the initial treatment block, despite being assigned a synchronous treatment onset date within the binned specification. While the scope for structural bias is minor, as these timing differences exclusively impact the initial treatment period for a subset of treated institutions, we address this explicitly by accounting for the underlying staggered adoption framework and potential treatment-effect heterogeneity across historical cohorts. To this end, we re-estimate Equation (2) utilizing calendar-month bins paired with the interaction-weighted cohort estimator proposed by Sun and Abraham (2021), with never-treated university clinics serving as the baseline comparison cohort. The resulting dynamic paths for admissions and mortality closely mirror our baseline estimates (Figure A.3), corroborating our main findings and indicating that the coarser treatment timing in the baseline specification does not materially affect our conclusions.

6 Conclusion

Labour disruptions within essential public services, including transportation, law enforcement, and medical care, have escalated globally, driven by fiscal austerity, structural rationalization,

²⁰We use all other hospitals in Hamburg and Berlin, who participated neither in the federal nor in the municipality strike, as alternative controls. Note that this alternative control group includes all hospitals in the two federal states, not only university hospitals, and is therefore potentially less comparable with respect to a range of factors, including size and scope of services provided.

and public sector welfare retrenchment. In the healthcare sector, these compounding pressures have triggered unprecedented waves of industrial action, generating severe concerns regarding restricted patient access and degraded clinical quality. Given the high-stakes nature of medical delivery, systematically evaluating the societal and individual costs of healthcare strikes represents a critical imperative for economic policy.

This paper leverages an unexpected nationwide physician strike in German university hospitals in 2006 to identify its causal effect on inpatient mortality. Utilizing a difference-in-differences design, we contrast dynamic changes in clinical outcomes between strike-participating tertiary institutions and control facilities that were legally prohibited from striking. Our empirical results show that the supply-side shock induced a 12% contraction in daily hospital admissions and a concurrent 9% surge in the raw in-hospital mortality rate at affected clinics. Adjusting for high-dimensional patient characteristics via a double-selection LASSO framework attenuates this mortality effect by half. This structural shift suggests that the raw mortality surge was driven jointly by strategic triage, where sicker, higher-acuity individuals were prioritized for admission, and genuine reductions in care quality. Conversely, we find no evidence of post-strike catch-up admissions or elevated out-of-hospital population mortality, alongside modest demand-side spillovers to neighbouring community hospitals that successfully absorbed displaced patients without experiencing elevated mortality.

Our study contributes with nuance to the sparse literature on healthcare labour disputes by isolating the impacts of a protracted, continuous supply shock, shifting the analytical focus beyond the brief, recurring disruptions examined in prior research. We interpret these findings as evidence that the operational efficacy and safety of essential services under labour stress depend fundamentally on the duration and scope of the structural disruption. A remaining limitation of our analysis is that inpatient mortality represents a rare and extreme clinical endpoint. It is highly plausible that non-lethal indicators of quality, such as readmission rates, clinical complications, or patient satisfaction, were similarly or more severely impacted. Furthermore, a comprehensive evaluation of the strike's broader economic externalities, including hospital financial deficits and the psychological workload borne by non-striking

personnel, remains outside the scope of this study. Given the significant costs we document, mapping the aggregate impacts of healthcare labour disputes across a broader matrix of social, financial, and clinical indicators represents a vital avenue for future research.

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Primum Non Nocere? Mortality Effects of a Physician
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— Online Appendix —

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Appendix A

TABLE A.1. Hospital Staffing and Workforce Allocation in Germany (2000–2010)

Year	All Hospital Facilities			University Medical Centres		
	Total Staff	Physicians	Other Medical	Total Staff	Physicians	Other Medical
2000	1.100,471	122.062	978.409	172.867	24.398	148.469
2001	1.101,356	123.819	977.537	173.114	24.758	148.356
2002	1.112,421	126.047	986.374	174.850	25.084	149.766
2003	1.096,420	128.853	967.567	173.091	25.154	147.937
2004	1.071,846	129.817	942.029	168.980	25.171	143.809
2005	1.063,154	131.115	932.039	170.160	25.435	144.725
2006	1.064,377	133.649	930.728	171.895	25.781	146.114
2007	1.067,287	136.267	931.020	172.782	26.241	146.541
2008	1.078,212	139.294	938.918	173.182	26.488	146.694
2009	1.096,520	143.967	952.553	177.535	27.632	149.903
2010	1.112,959	148.696	964.263	181.954	28.443	153.511

NOTE.— Own calculations based on data from the Information System of the Federal Health Monitoring (*Gesundheitsberichterstattung des Bundes*). The data reflects personnel headcounts over time across all domestic acute-care clinics and university hospitals.

TABLE A.2. Strike Timelines and Durations Across University Hospitals

University Hospital	District Code	Federal State	Start Date	End Date
Tübingen	416	Baden-Württemberg	20/03/2006	16/06/2006
Heidelberg	221	Baden-Württemberg	16/03/2006	16/06/2006
Ulm	421	Baden-Württemberg	20/03/2006	16/06/2006
Freiburg	311	Baden-Württemberg	16/03/2006	16/06/2006
München	162	Bayern	16/03/2006	16/06/2006
Erlangen-Nürnberg	562	Bayern	20/03/2006	16/06/2006
Würzburg	663	Bayern	16/03/2006	16/06/2006
Regensburg	362	Bayern	20/03/2006	16/06/2006
Greifswald	001	Mecklenburg-Vorpommern	19/04/2006	16/06/2006
Rostock	003	Mecklenburg-Vorpommern	19/04/2006	16/06/2006
Hannover	241	Niedersachsen	20/03/2006	16/06/2006
Göttingen	152	Niedersachsen	21/03/2006	16/06/2006
Essen	113	Nordrhein-Westfalen	16/03/2006	16/06/2006
Münster	515	Nordrhein-Westfalen	04/04/2006	16/06/2006
Aachen	313	Nordrhein-Westfalen	20/03/2006	16/06/2006
Köln	315	Nordrhein-Westfalen	20/03/2006	16/06/2006
Düsseldorf	111	Nordrhein-Westfalen	22/03/2006	16/06/2006
Bonn	314	Nordrhein-Westfalen	16/03/2006	16/06/2006
Mainz	315	Rheinland-Pfalz	16/03/2006	16/06/2006
Dresden	262	Sachsen	18/04/2006	16/06/2006
Leipzig	713	Sachsen	18/04/2006	16/06/2006
Magdeburg	003	Sachsen-Anhalt	21/03/2006	16/06/2006
Halle	202	Sachsen-Anhalt	16/03/2006	16/06/2006
Jena	053	Thüringen	24/03/2006	16/06/2006

NOTE.— Own calculations based on collected strike data. This table presents the precise calendar timeline for each of the 24 strike-participating university hospitals. District codes correspond to the official AGS (*Amtlicher Gemeindeschlüssel*) regional metrics. The uniform end date of June 16, 2006, reflects the signing of the landmark national collective agreement (*TV-Ärzte*) between the *Marburger Bund* and the employers' association (*TdL*).

TABLE A.3. Variable Definitions

Variable Label	Data Type	Operational Definition
Treatment Indicator	Binary (0/1)	Takes the value of 1 for patient admissions to a strike-participating university hospital during the active strike window, and 0 otherwise.
Female	Binary (0/1)	Indicator variable tracking patient gender, equal to 1 for female individuals.
Age	Continuous	Exact patient chronological age recorded in years at the time of clinical admission.
Married	Binary (0/1)	Indicator variable tracking civil status, equal to 1 for married individuals.
Daily Admissions (cases)	Integer	Aggregate absolute count of hospitalizations per day within a specific facility.
Log Daily Admissions	Continuous	Natural logarithm of the absolute daily volume of admissions per facility.
Length of Stay	Integer	Total duration of individual inpatient hospitalization operationalised in days.
Surgery Rate	Continuous	Number of operative surgical interventions performed per 100 patient admissions.
In-Hospital Mortality Rate	Continuous	Total number of inpatient fatalities per 100 documented hospital admissions.
10-Day Mortality Rate	Continuous	Inpatient deaths occurring within a strict 10-day post-admission window per 100 admissions.
District Mortality Rate	Continuous	Population-adjusted regional macro mortality expressed as deaths per 10,000 inhabitants.
ACS Admission Rate	Continuous	Ambulatory Care Sensitive admissions per 100 facility hospitalizations, constructed using the consensus clinical guidelines of Sundmacher et al. (2015) .
Clinical Urgency Index	Continuous	Granular diagnostic severity metric bounded between 0 and 1, computed following the empirical machine-learning classification framework of Krämer, Schreyögg and Busse (2019) .
Emergency Admission Rate	Continuous	Acute admissions per 100 facility hospitalizations; mapping an emergency indicator equal to 1 if the underlying clinical urgency index exceeds the structural threshold of 0.5 (Krämer, Schreyögg and Busse, 2019).
Deferrability Index	Continuous	Continuous institutional metric bounded between 0 and 1, operationalized as the ratio of weekend admissions to week-day admissions mapped per unique 3-digit ICD-10 diagnosis code (Card, Dobkin and Maestas, 2009 , Halla, Liu and Liu, 2019).
Log Regional Deaths	Continuous	Natural logarithm of the absolute daily volume of structural deaths aggregated at the administrative district (<i>Kreis</i>) level.

TABLE A.4. Difference-in-Differences Balance Tests on Hospital-Level Characteristics

Dependent Variable	Observations	Baseline Mean	(1)	(2)	(3)
Number of Beds	395	1130.48	432.597** (191.028)	−22.986 (27.176)	−5.064 (15.843)
ICU Bed Share	395	0.08	0.020** (0.009)	0.001 (0.002)	0.002 (0.003)
Total Physicians	395	573.50	337.988*** (106.905)	37.294 (23.928)	8.034 (20.568)
Female Physician Share	395	0.33	−0.003 (0.017)	0.001 (0.008)	−0.004 (0.011)
Physicians in Charge	395	40.74	15.316 (12.246)	−2.302 (4.399)	−3.242 (3.192)
Assistant Physicians	395	433.12	227.567** (87.075)	20.984 (17.948)	10.834 (11.049)
Non-Physician Staff	395	3294.67	1972.984*** (560.346)	202.625* (108.468)	44.069 (104.998)
Female Non-Physician Share	395	0.77	−0.010 (0.013)	−0.008* (0.004)	−0.002 (0.003)
Trainee Positions	395	352.22	134.706** (66.874)	−20.573 (19.103)	−22.718 (17.253)
Annual Childbirths	395	1167.57	508.146 (413.760)	51.139 (50.383)	9.083 (38.120)
C-Section Share	395	0.28	0.119* (0.059)	−0.005 (0.023)	−0.002 (0.023)
Calendar Year Fixed Effects			✓	✓	✓
Hospital Fixed Effects				✓	✓
Hospital Linear Trends					✓

NOTE.— Own calculations based on data from [FDZ der Statistischen Ämter des Bundes und der Länder \(2008\)](#). Robust standard errors clustered at the hospital level are reported in parentheses. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Regressions successively include calendar year fixed effects, hospital fixed effects, and hospital-specific linear time trends across columns (1) to (3). In contrast to the daily micro data main specifications, treatment here is operationalized at the annual level for the calendar year 2006 due to the reporting frequency of institutional characteristics. A zero share of C-sections is assumed for facilities without active maternity wards. Comprehensive variable definitions are detailed in [Table A.3](#) of [Appendix A](#).

TABLE A.5. Difference-in-Differences Estimates: Heterogeneity Analysis by Deferability

Dependent Variable	Quartile 1 (Least Deferrable)			Quartile 2		
	Mean	(1)	(2)	Mean	(1)	(2)
Daily Admissions (cases)	167.82	−22.039*** (4.660)	−19.756*** (2.735)	166.81	−22.870*** (4.848)	−20.542*** (2.758)
Log Daily Admissions	3.64	−0.160*** (0.025)	−0.154*** (0.022)	3.58	−0.173*** (0.024)	−0.167*** (0.022)
In-Hospital Mortality Rate (%)	1.53	0.186** (0.091)	0.196** (0.092)	1.52	0.313*** (0.089)	0.312*** (0.090)
10-Day Mortality Rate (%)	0.66	0.093 (0.058)	0.091 (0.058)	0.66	0.126** (0.062)	0.121* (0.062)
Log Daily Inpatient Deaths	0.24	−0.000 (0.018)	−0.000 (0.017)	0.24	−0.034* (0.020)	−0.028 (0.017)
Length of Stay (days)	7.48	0.279*** (0.098)	0.300*** (0.087)	8.79	0.477*** (0.084)	0.459*** (0.081)
Surgery Rate (%)	38.34	−0.398 (1.519)	−2.065* (1.175)	42.73	0.184 (1.900)	−1.780 (1.201)
Dependent Variable	Quartile 3			Quartile 4 (Most Deferrable)		
	Mean	(1)	(2)	Mean	(1)	(2)
Daily Admissions (cases)	155.72	−20.668*** (4.449)	−18.715*** (2.556)	142.07	−18.800*** (4.659)	−16.517*** (2.351)
Log Daily Admissions	3.49	−0.127*** (0.024)	−0.128*** (0.018)	3.41	−0.044*** (0.015)	−0.040*** (0.010)
In-Hospital Mortality Rate (%)	1.30	0.165*** (0.046)	0.168*** (0.043)	2.79	−0.007 (0.095)	−0.010 (0.092)
10-Day Mortality Rate (%)	0.69	0.041 (0.037)	0.040 (0.036)	1.91	0.001 (0.072)	−0.006 (0.072)
Log Daily Inpatient Deaths	0.18	−0.004 (0.015)	−0.001 (0.014)	0.35	−0.021 (0.018)	−0.017 (0.013)
Length of Stay (days)	9.05	0.407*** (0.104)	0.383*** (0.091)	7.55	0.116 (0.075)	0.118* (0.063)
Surgery Rate (%)	35.29	0.493 (1.426)	−1.161 (0.884)	27.83	1.030 (1.252)	−0.800 (0.806)
Calendar Year Fixed Effects		✓	✓		✓	✓
Hospital Fixed Effects		✓	✓		✓	✓
Hospital Linear Trends			✓			✓

NOTE.— Own calculations based on data from [FDZ der Statistischen Ämter des Bundes und der Länder \(2008\)](#). Robust standard errors clustered at the hospital level are reported in parentheses. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. All regressions condition on hospital fixed effects, day-of-the-week fixed effects, week-of-the-year fixed effects, calendar year fixed effects, and hospital-specific linear time trends where indicated. Regressions are weighted by the baseline average volume of pre-strike admissions per hospital. The clinical deferability index is constructed following the frameworks of [Card, Dobkin and Maestas \(2009\)](#) and [Halla, Liu and Liu \(2019\)](#). Comprehensive variable definitions are detailed in [Table A.3 of Appendix A](#).

TABLE A.6. Difference-in-Differences Placebo Regressions Using Backdated 2004 Timelines

Dependent Variable	Observations	Baseline Mean	(1)	(2)
Daily Admissions (cases)	143,744	158.32	−1.032 (4.081)	−1.311 (4.265)
Log Daily Admissions	143,744	4.90	0.009 (0.014)	0.007 (0.015)
In-Hospital Mortality Rate (%)	143,744	1.75	0.021 (0.058)	−0.006 (0.038)
10-Day Mortality Rate (%)	143,744	0.96	0.002 (0.037)	−0.015 (0.025)
Daily Number of Deaths	143,744	2.46	0.002 (0.110)	−0.034 (0.101)
Length of Stay (days)	143,744	8.17	−0.021 (0.061)	−0.026 (0.062)
Surgery Rate (%)	143,744	35.95	2.610 (1.853)	2.395 (1.865)
ACS Admission Rate (%)	143,744	20.30	0.056 (0.178)	0.077 (0.180)
Emergency Admission Rate (%)	143,156	28.41	−0.121 (0.204)	−0.106 (0.202)
Clinical Urgency Index	143,156	0.34	−0.002 (0.002)	−0.001 (0.002)
Deferrability Index	143,549	0.12	−0.000 (0.000)	−0.000 (0.000)
Calendar Year Fixed Effects			✓	✓
Hospital Fixed Effects			✓	✓
Hospital Linear Trends				✓

NOTE.— Own calculations based on data from [FDZ der Statistischen Ämter des Bundes und der Länder \(2008\)](#). Robust standard errors clustered at the hospital level are reported in parentheses. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. All regression specifications include hospital fixed effects, day-of-the-week fixed effects, week-of-the-year fixed effects, calendar year fixed effects, and hospital-specific linear time trends where indicated. Regressions are weighted by the baseline average volume of pre-strike admissions per hospital. Estimates evaluate a counterfactual placebo shock backdated to the identical calendar window in 2004, a period marked by zero actual industrial dispute activity. Comprehensive variable definitions are detailed in [Table A.3 of Appendix A](#).

TABLE A.7. Difference-in-Differences Robustness Estimates: Alternative Control Group

Dependent Variable	Observations	Baseline Mean	(1)	(2)
Daily Admissions (cases)	267,382	161.83	−21.977*** (3.718)	−19.536*** (2.354)
Log Daily Admissions	267,382	4.72	−0.138*** (0.016)	−0.125*** (0.013)
In-Hospital Mortality Rate (%)	267,382	2.07	0.270*** (0.045)	0.199*** (0.041)
10-Day Mortality Rate (%)	267,382	1.19	0.135*** (0.036)	0.095*** (0.034)
Daily Number of Deaths	267,382	3.07	0.006 (0.090)	−0.035 (0.055)
Length of Stay (days)	267,382	8.28	0.392*** (0.064)	0.261*** (0.053)
Surgery Rate (%)	267,382	37.35	−1.765 (1.607)	−3.004** (1.170)
ACS Admission Rate (%)	267,382	21.93	−0.397* (0.216)	−0.496*** (0.175)
Emergency Admission Rate (%)	84,648	27.65	1.494*** (0.350)	1.484*** (0.353)
Clinical Urgency Index	266,644	0.37	0.017*** (0.003)	0.014*** (0.002)
Deferrability Index	266,935	0.13	0.004*** (0.001)	0.004*** (0.001)
Calendar Year Fixed Effects			✓	✓
Hospital Fixed Effects			✓	✓
Hospital-Specific Linear Trends				✓

NOTE.— Own calculations based on data from [FDZ der Statistischen Ämter des Bundes und der Länder \(2008\)](#). Robust standard errors clustered at the hospital level are reported in parentheses. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. All regression specifications include hospital fixed effects, day-of-the-week fixed effects, week-of-the-year fixed effects, calendar year fixed effects, and hospital-specific linear time trends where indicated. Regressions are weighted by the baseline average volume of pre-strike admissions per hospital. The alternative control group consists of all non-striking hospitals in Berlin and Hesse. Comprehensive variable definitions are detailed in [Table A.3](#) of [Appendix A](#).

TABLE A.8. Difference-in-Differences Robustness Estimates: Alternative Treatment Indicator

Dependent Variable	Observations	Baseline Mean	(1)	(2)
Daily Admissions (cases)	143,744	158.32	−20.065*** (4.984)	−17.453*** (2.645)
Log Daily Admissions	143,744	4.90	−0.107*** (0.019)	−0.103*** (0.014)
In-Hospital Mortality Rate (%)	143,744	1.75	0.128*** (0.044)	0.134*** (0.042)
10-Day Mortality Rate (%)	143,744	0.96	0.055 (0.036)	0.054 (0.035)
Daily Number of Deaths	143,744	2.46	−0.097 (0.095)	−0.064 (0.062)
Length of Stay (days)	143,744	8.17	0.226*** (0.058)	0.217*** (0.054)
Surgery Rate (%)	143,744	35.95	−0.473 (1.510)	−1.856* (0.945)
ACS Admission Rate (%)	143,744	20.30	0.009 (0.233)	−0.134 (0.193)
Emergency Admission Rate (%)	143,156	28.41	1.364*** (0.334)	1.312*** (0.329)
Clinical Urgency Index	143,156	0.34	0.010*** (0.002)	0.010*** (0.002)
Deferrability Index	143,549	0.12	0.003*** (0.001)	0.003*** (0.001)
Calendar Year Fixed Effects			✓	✓
Hospital Fixed Effects			✓	✓
Hospital Linear Trends				✓

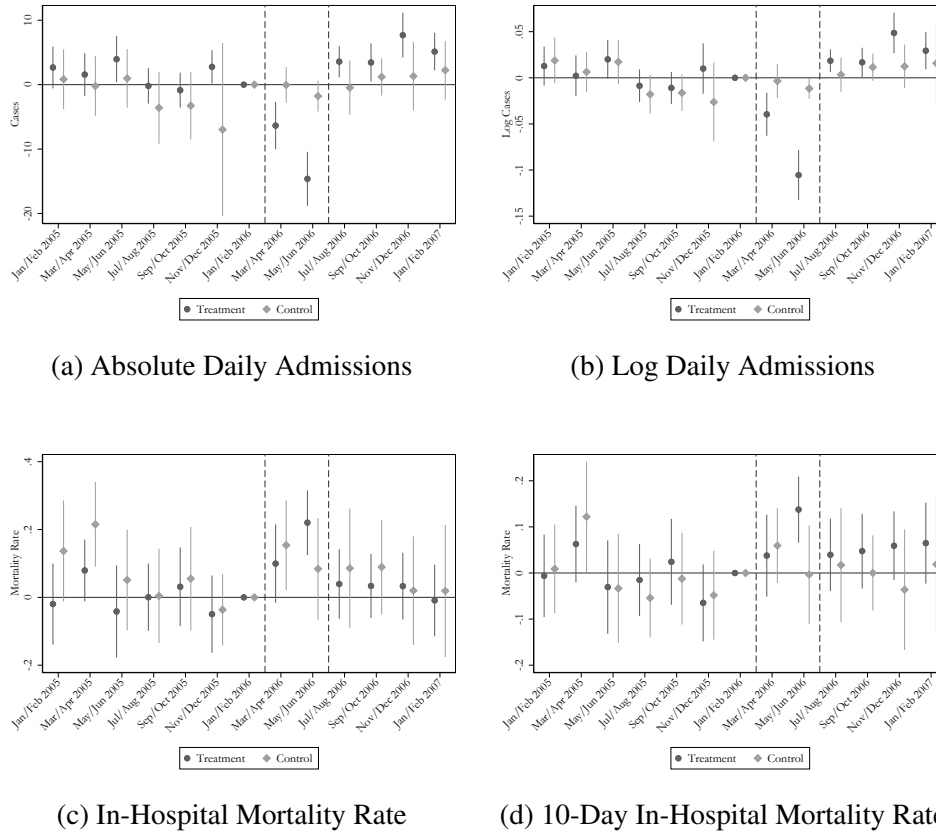
NOTE.— Own calculations based on data from [FDZ der Statistischen Ämter des Bundes und der Länder \(2008\)](#). Robust standard errors clustered at the hospital level are reported in parentheses. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. All regression specifications include hospital fixed effects, day-of-the-week fixed effects, week-of-the-year fixed effects, calendar year fixed effects, and hospital-specific linear time trends where indicated. Regressions are weighted by the baseline average volume of pre-strike admissions per hospital. The alternative treatment indicator tracks the daily stock proportion of active inpatients whose institutionalisation overlapped with an active strike day at any point during their stay. Comprehensive variable definitions are detailed in [Table A.3 of Appendix A](#).

TABLE A.9. Formal Identification Test: Baseline Pre-Dispute Trend Differences

Independent Variable	(1) Log Daily Admissions	(2) In-Hospital Mortality (%)	(3) 10-Day Mortality (%)	(4) Absolute Daily Admissions
Linear Trend $\times$ Treated	-0.0000262 (0.0000325)	-0.0000480 (0.0000448)	-0.0000434 (0.0000264)	-0.00972 (0.0106)

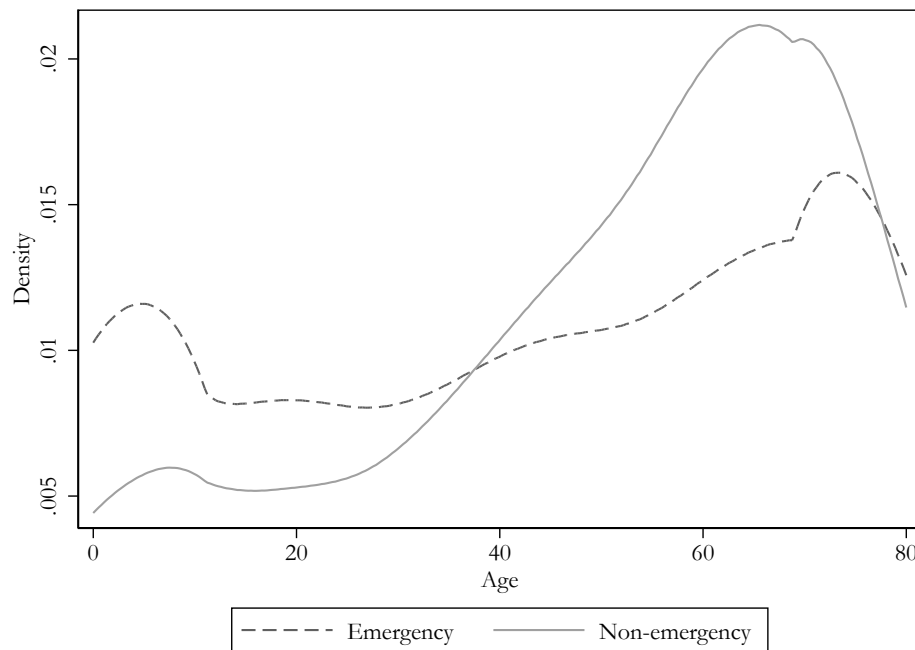
NOTE.— Own calculations based on data from [FDZ der Statistischen Ämter des Bundes und der Länder \(2008\)](#). Robust standard errors clustered at the hospital level are reported in parentheses. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Regressions are weighted by the baseline average volume of pre-strike admissions per hospital. This table reports coefficient estimates from a falsification specification testing for differential parametric trends between the 24 strike-participating university hospitals and the 9 structurally insulated control facilities during the entire pre-strike baseline window. Comprehensive variable definitions are detailed in [Table A.3](#) of [Appendix A](#).

FIGURE A.1. Event Study Estimates: Daily Admissions and Inpatient Mortality by Hospital Treatment Status



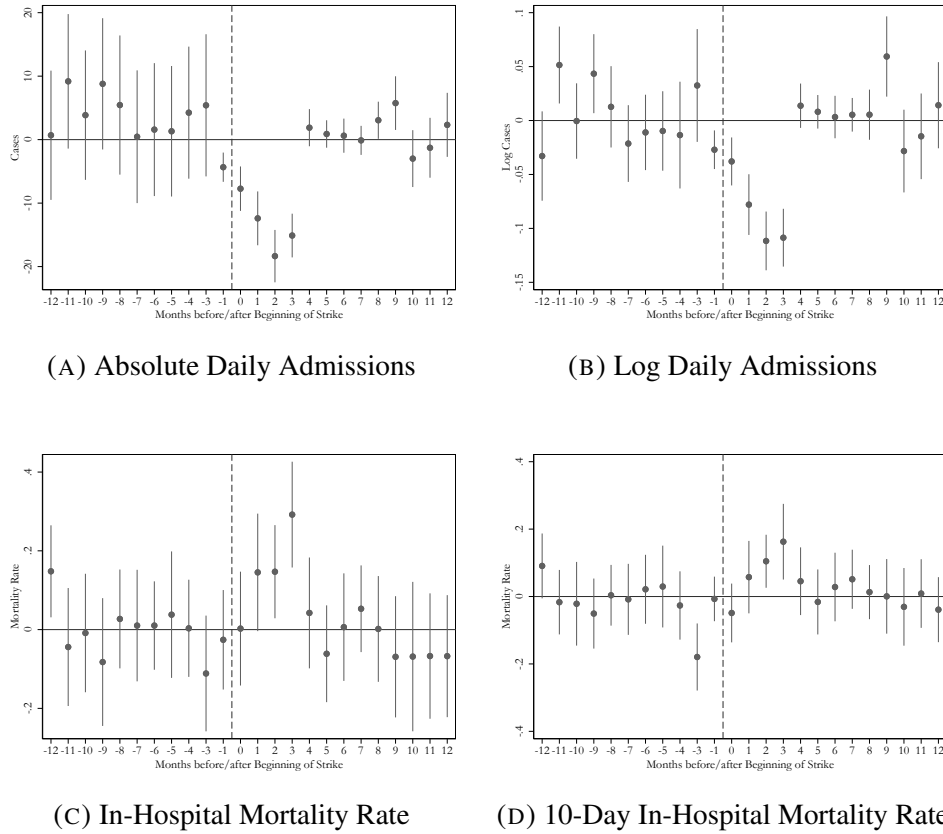
NOTE.— Own calculations based on data from [FDZ der Statistischen Ämter des Bundes und der Länder \(2008\)](#). The panels plot conditional descriptive trajectories for absolute daily admissions (Panel a), log daily admissions (Panel b), the global in-hospital mortality rate (Panel c), and the restricted 10-day in-hospital mortality rate (Panel d), contrasting the 24 strike-participating university hospitals (treatment group) against the 9 structurally insulated facilities (control group). All underlying regression specifications condition on hospital fixed effects, day-of-the-week fixed effects, and week-of-the-year fixed effects. The vertical dashed lines demarcate the initiation and formal resolution of the universal strike period. Shaded areas (or error bars) represent 95% confidence intervals computed from robust standard errors clustered at the hospital level. Regressions are weighted by the baseline average volume of pre-strike admissions per hospital. Comprehensive variable definitions are detailed in [Table A.3 of Appendix A](#).

FIGURE A.2. Baseline Patient Age Distributions by Admission Emergency Status



NOTE.— Own calculations based on data from [FDZ der Statistischen Ämter des Bundes und der Länder \(2008\)](#). The figure plots the empirical kernel density distributions of patient age at the individual admission level, stratified by clinical emergency status. The underlying microdata includes acute inpatient admissions across all 33 sample university hospitals. As documented, the emergency cohort exhibits a systematically younger distribution than elective patient groups, highlighting why adjusting for dynamic case-mix transformations is essential for our identification framework. Comprehensive variable definitions are detailed in [Table A.3](#) of [Appendix A](#).

FIGURE A.3. Staggered-Adoption Interaction-Weighted Event Study Estimates



NOTE.— Own calculations based on data from [FDZ der Statistischen Ämter des Bundes und der Länder \(2008\)](#). The panels plot dynamic treatment effects estimated via the interaction-weighted cohort estimator proposed by [Sun and Abraham \(2021\)](#) to account for staggered treatment adoption and heterogeneous treatment effects. Panels (a) and (b) trace the trajectories for absolute and log daily admissions; Panels (c) and (d) trace the trajectories for global and 10-day in-hospital mortality rates. All underlying regression specifications condition on hospital and 10-day in-hospital mortality rates. All underlying regression specifications condition on hospital fixed effects, day-of-the-week fixed effects, and week-of-the-year fixed effects, using the period $t - 2$ as the baseline reference category. The vertical dashed lines demarcate the universal active strike window. Shaded areas represent 95% confidence intervals computed from robust standard errors clustered at the hospital level, and regressions are weighted by the baseline average volume of pre-strike admissions per hospital. Comprehensive variable definitions are detailed in [Table A.3 of Appendix A](#).